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Order batching in an automated warehouse with several vertical lift modules: Optimization and experiments with real data

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ABSTRACT

The vertical lift module is an automated storage and retrieval system widely used in warehouses. The performance of a warehouse with vertical lift modules is highly correlated with the efficiency of the order picking. Order batching, namely regrouping customers' orders into batches to be collected from the module, constitutes a critical decision impacting the picking efficiency. In this paper, we provide optimization models for order batching, with the objective of minimizing total completion time, that is, the time required to collect a given set of customers' orders. We first consider the case of one vertical lift module and then extend our approach to study a warehouse with several modules. We use real data from two companies operating in different sectors in order to test and validate our models. Numerical experiments show that our models perform much better than the batching method currently used by these companies. For complex cases that cannot be solved within a reasonable timeframe with Cplex, we develop a metaheuristic approach, which generally yields very good solutions in less than one minute. This paper investigates problems that are firmly grounded in practice. Our batching models and metaheuristic approach have been implemented in practice and are currently used by some companies.

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1. Introduction

1.1. The vertical lift module as one of the main automated picking systems

Automated Storage and Retrieval Systems (AS/RS) are increasingly used in the warehouses of different companies. These systems offer several advantages, such as space savings and increased productivity. Utilization of AS/RS implies that the machine brings the products to the operator (picker), who stays in front of the machine and does not need to move from one storage location to another. In the literature, such a system is referred to as a part to picker system as opposed to a picker to part system, where pickers have to retrieve products by walking down aisles until they reach the location where the product is stored. The Vertical Lift Module (VLM) is a widely used type of AS/RS, which comprises a set of insertion/extraction storing trays, on which products are stored (see Fig. 1 for an illustration). In this case, the picking area, also called pod, typically consists of one or more pickers, who retrieve products from one or more VLMs.

1.2. Picking process with VLMs

Order picking can be defined as the process of retrieving products from their storage location to execute customers' orders. A customer's order is represented by a set of lines, where each line corresponds to a required quantity of a given product. Each product is stored on a given tray of a given VLM. Once a set of orders is received, the first decision is how to regroup these orders into different batches. A batch is composed of a subset of orders and, consequently, of the set of lines corresponding to these orders. In practice, a given order cannot be split between different batches. In other words, all the lines of a given order must be placed in the same batch. Then, after the batching decision, the batches have to be picked from the pod (i.e., the picking area with VLMs). Based on our observation and study of VLM systems in different warehouses, we hereafter describe the picking process with VLMs.

For each batch, the picker receives a number of bins, which is equal to the number of orders included in the batch; each order must be placed in a different bin. Each time the picker retrieves an order line (i.e., a certain quantity of a given product) from a VLM

Abbreviations: AS/RS, Automated Storage and Retrieval Systems; VLM, Vertical Lift Module; AVS/RS, Autonomous Vehicle Storage and Retrieval Systems; SBS/RS, Shuttle Based Storage and Retrieval Systems; SA, Simulated Annealing; MFV, Mean Fitness Value; ER, Efficiency Rate; IR, Improvement Rate.

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List of symbols

Symbol	Meaning
B	Number of batches to be created.
$\mathcal{B} = \{1, \ldots, B\}$	Set of all batches.
C	Total completion time for all batches.
C^b	Completion time associated with batch b.
C^b_l	Completion time for the l-th tray associated with batch b.
K	Total number of trays in the VLM.
$\mathcal{K} = \{1, \ldots, K\}$	Set of trays in the VLM.
K^b	Number of trays that have to be visited for batch b.
$Kb = \{1, \ldots, K^b\}$	Set of trays that have to be visited for batch b.
K_v	Total number of trays in VLM v.
$\mathcal{K}_v = \{1, \ldots, K_v\}$	Set of trays of VLM v.
K^b_v	Number of trays that have to be visited on VLM v for batch b.
$\mathcal{K}b_v = \{1, \ldots, K^b_v\}$	Set of trays that have to be visited on VLM v for batch b.
L^b_l	Number of order lines that have to be collected on the l-th tray visited for batch b.
L^b_{lv}	Number of order lines that have to be collected on the l-th tray of VLM v for batch b.
N	Number of considered customers' orders.
$\mathbb{N} = \{1, \ldots, N\}$	Set of considered customers' orders.
p	Picking time of one order line.
P	Total picking time.
P^b	Picking time for batch b.
P^b_l	Time required to collect items from the l-th tray for batch b.
P^b_{lv}	Time required to collect items from the l-th tray of VLM v for batch b.
r	Time required by the VLM to store the current tray and to extract the next one.
S	Batch size (i.e., the maximum number of orders to put in a batch).
$T_{k\ b}$	Binary variable indicating whether tray k is visited by batch b or not.
T_{kvb}	Binary variable indicating whether tray k of VLM v is visited by batch b or not.
U_{kn}	Binary parameter indicating whether tray k is visited by order n or not.
U_{kvn}	Binary parameter indicating whether tray k of VLM v is visited by order n or not.
V	Total number of VLMs.
$\vartheta = \{1, \ldots, V\}$	Set of all VLMs.
W	Total waiting time of the picker.
W^b	Waiting time associated with batch b.
W^b_l	Waiting time of the picker in front of the l-th tray that has to be visited for batch b.
W^b_{lv}	Waiting time of the picker in front of the l-th tray that has to be visited by VLM v for batch b.
x_{nb}	Binary variable indicating whether order n is included in batch b or not.

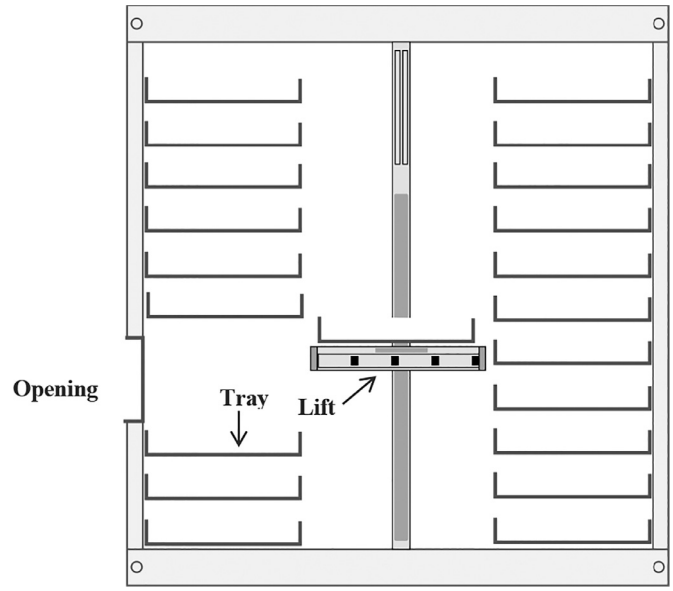

Fig. 1. VLM illustration (side view).

tray, he places the product in the bin associated with the order requiring this product. The picker begins by collecting the order lines from the first tray of the first VLM that has to be visited and continues until no other order lines have to be retrieved from this tray. The current tray is then returned to its location and the next tray is extracted. Meanwhile, the picker walks to the next VLM and collects items from the first tray that has to be visited on this second VLM. The picker continues moving from one VLM to another until he finishes collecting items from the first tray of each VLM (in other words, until all VLMs containing products that have to be picked are visited once). The picker then returns to the first VLM and repeats the process with the second tray that has to be visited on this VLM. The process continues as described above until all order lines belonging to the current batch are collected from the different VLMs. The bins leave the pod as soon as all items are collected, and new bins, associated with a new batch, are received by the picker. The picker repeats this process until all batches are picked. Therefore, the completion time for retrieving all batches will only depend on the picking time for collecting all order lines and the waiting time of the picker in front of the machines. The picking process described above is illustrated in Fig. 2. (where V is the number of VLMs and Tb the number of picking rounds for batch b).

1.3. Important role of the batching decision in a VLM system

Order picking is one of the operations with the most impact on a warehouse's overall performance. It is the most labor-intensive operation in manual warehouses and represents approximately 55% of all operating costs (Tompkins, White, Bozer, & Tanchoco, 2010), and a primordial operation in warehouses equipped with automated systems (De Koster, Le-Duc, & Roodbergen, 2007). Achieving a higher throughput (i.e., the number of orders collected by a picker within a certain time period) is currently one of the main objectives of warehouse managers (Yu & De Koster, 2010). Improving throughput is highly correlated with the batching decision, or the decision on regrouping customers' orders into batches. According to Pazour and Meller (2013), batching offers many advantages and plays an important role in VLM systems. For instance, when products from several orders have to be retrieved from the same tray of a given VLM, it is more efficient to retrieve them

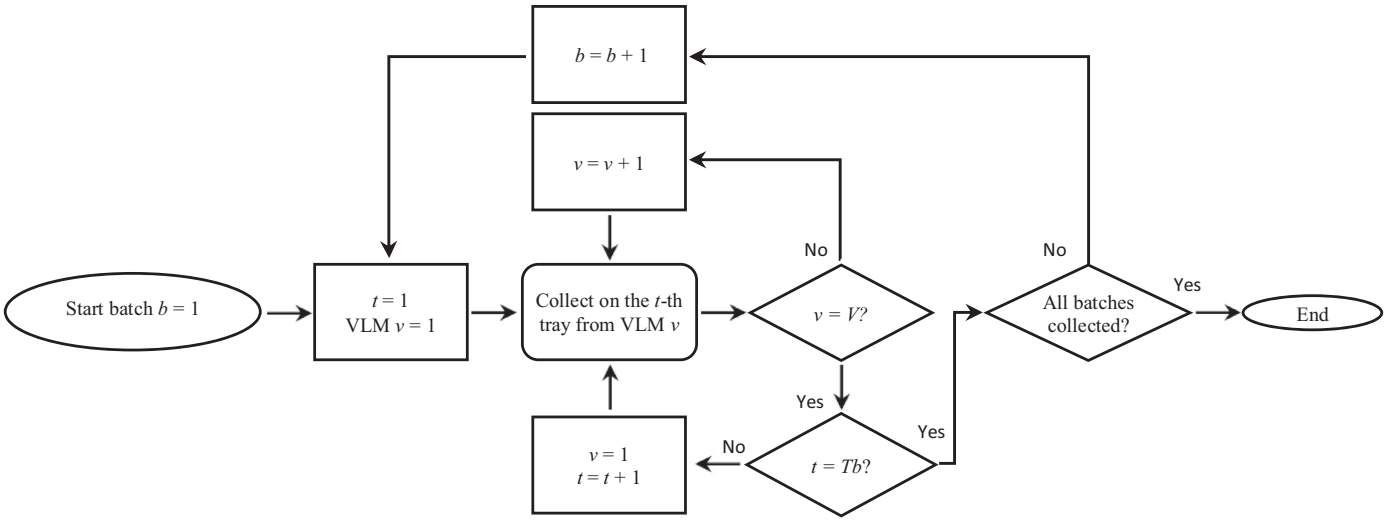

Fig. 2. Order picking with VLMs.

simultaneously instead of having to call this tray several times. Indeed, this leads to time and cost savings.

Note that two situations should be distinguished with regard to order batching: "Offline" batching, when all orders are known in advance and therefore are all available for batching, and "Online" batching, when only some orders are available from the beginning, and the arrival time of orders needs to be considered during the planning period. Our study deals with offline batching, which is the approach most commonly used in practice.

1.4. Motivation for our research

The order batching decision is made by most companies using VLMs. However, we observed that many of these companies use basic batching procedures, such as including orders in a batch according to their arrival date or their due date until the batch is full. We were asked by practitioners to investigate whether a more sophisticated batching strategy could lead to a shorter completion time (i.e., the time required to collect a set of customers' orders from one or more VLMs). In spite of the critical impact of batching on order picking performance, the literature has not paid adequate attention to this decision problem in the case of warehouses using VLMs. Indeed, most works addressing order batching and picking problems in the archives of the literature focus on picker to part systems (manual picking systems).

Thus, on the one hand, our research is motivated by the real need to develop batching decision models for companies using VLMs and by the potential gain that such models can bring in practice (we will prove in this paper that our proposed batching models can lead to significant gains for companies). On the other hand, this work is motivated by the lack of research on the batching problem for automated warehouses in general, and for VLMs in particular. Finally, we consider the complex case of several VLMs, for which synchronization of picking operations between the different VLMs plays a crucial role in optimizing the total completion time, but which is very challenging to take into account.

1.5. Contribution of our research

This paper contributes to the literature and practice by providing batching optimization models for warehousing systems operating VLMs. The objective of our models is to minimize the total completion time required to collect a set of customers' orders from one or more VLMs, which is in line with the objectives of warehouse managers in practice. Total completion time is the sum of the effective picking time and the waiting time. To calculate picking time, we consider two situations: the picking time per tray is either assumed to be negligible or proportional to the number of lines that have to be collected on the tray.

We first consider the case of one VLM and then extend our work to the more complex case of multiple VLMs. In each case, we start by evaluating the completion time of a given set of customers' orders. As will be explained later in the paper, this important step will help us identify a much simpler objective function for our optimization models than the complex and non-linear completion time function. In most cases, we prove that this new objective is equivalent to minimizing completion time. We then develop our models for optimization of the batching strategy for warehouses with VLMs. To test and validate our models, we use real data extracted from the databases of two companies operating in different sectors. This will show the gain our models can offer in practice and will lead to some interesting insights. For more complex instances that cannot be efficiently solved with commercial optimization software, we finally develop a metaheuristic approach in order to meet the practical requirements in terms of the maximum acceptable computational time. We conduct experiments and show that our metaheuristic approach is able to find very good solutions in less than one minute, and it performs much better than the approaches currently used by companies.

The rest of this paper is organized as follows: In Section 2, we review the relevant literature. We dedicate Section 3 to developing the optimization model and experiments associated with the case of one VLM. In Section 4, we focus on the case of multiple VLMs. Section 5 presents the proposed metaheuristic approach for order batching and discusses its performance. We finally conclude and give future work directions.

2. Literature review

Our study brings together two streams of literature. First, it is linked to the research conducted on the performance evaluation of VLM systems in particular, and to AS/RS in general. Second, our work contributes to the literature dealing with order batching decisions in warehousing. In this section, we review each of these research streams.

2.1. Automated storage and retrieval technologies

Thanks to the rapid development of automated technologies, different types of AS/RS have emerged. AS/RS widely used in

practice include VLM systems, carousel systems, Shuttle Based Storage and Retrieval Systems (SBS/RS) and Autonomous Vehicle Storage and Retrieval Systems (AVS/RS). Each of these may also have different variants. Whereas most of these technologies share the same basic operating characteristics, they also have some technical differences that lead to different assumptions and analytical difficulties in terms of the evaluation of their performance. In the following sections, we explain the main differences between these AS/RS.

In VLM systems, products are stored on front and rear tray locations. When a tray is requested, an extractor travels vertically between the two columns of trays, pulls the requested tray from its location and brings it to the picker. In carousel systems, trays are linked by a chain rotating horizontally (generally in both directions) and revolve on an oval track. Thus, the time required to release a tray from a carousel is highly correlated with its original position and its destination point (basically, the number of trays in-between) and, consequently, with the retrieval sequence of the different items. However, in a VLM, the analogous time required is roughly the same for all trays, since the vertical moving speed of the VLM is much higher than the horizontal speed.

In AVS/RS, the main component is an autonomous vehicle moving horizontally to search or store the unit load. At the end of the rail on which the vehicle is placed, lifts allow vertical movement (Ekren & Heragu, 2011). Thus, the vehicles in the system travel between fixed storage shelves to deposit or retrieve items, and the lifts are able to access to the necessary height to reach the load, which is then taken by the vehicle to the required location. The SBS/RS can be considered as a variant of AVS/RS and usually works with aisle and tier captive shuttles instead of vehicles (Lerher, Ekren, Sari, & Rosi, 2016). Therefore, in this case, the horizontal movement is undertaken by shuttle carriers instead of an autonomous vehicle.

VLMs and carousels are generally grouped in multiple units. For instance, the companies that we have studied have between 1 and 4 VLMs in their different warehouses. AVS/RS and SBS/RS belong to the category of fixed aisle systems, which are much bigger than VLMs/carousels and are generally used individually. When designing warehouses with AVS/RS or SBS/RS, one of the main questions is how to determine the number of aisles and the number of horizontal locations per aisle. Configuration of warehouses with VLMs or carousels raises other questions, such as the number of units and the number of trays per unit. It is important to note that having more than one unit saves time, as carousels/VLMs have more possibility of working under masked time.

For a comprehensive description of AS/RS, we refer the reader to the works published by Manzini, Gamberi, and Regattieri (2006) and Roodbergen and Vis (2009).

2.1.1. VLM

In the literature dedicated to AS/RS, much attention has been paid to warehouses with carousel systems and AVS/RS, whereas only few works focus on VLM systems. In papers dealing with VLMs, the main objective usually was to determine system throughput and to measure picker utilization. For instance, in Meller and Klote (2004), analytical expressions of throughput for one or more VLMs were obtained by using a two-server queuing model. The authors also determined picker utilization assuming a deterministic and exponential picking time distribution. Dukic, Opetuk, and Lerher (2015) used the same methodology for VLMs with a dual tray opening. This very particular type of VLMs has two trays that can be waiting at the opening point, which is a different technology from the one we studied. The method used for throughput calculation in this study is similar to that used in Meller and Klote (2004). Recently, Rosi, Grasic, Dukic, Opetuk, and Lerher (2016) investigated the advantages of VLM systems in terms of the reduction in mean cycle time or increased throughput. This was evaluated through simulation, comparing several VLM configurations.

2.1.2. Carousel

Carousel systems have been investigated by a large number of research studies, as reported by Litvak and Vlasiou (2010) in their overview of the literature. Given the wide variety of works on carousel systems, here we focus on some papers dealing with order picking, as they are more relevant to our research. Different picking strategies for carousels have been investigated in the literature. In the case of a single order, Litvak and Adan (2001) developed the nearest item heuristic, in which the carousel rotates toward the closest tray where an item is to be retrieved. In Litvak and Adan (2002), the m-step strategy was proposed. According to this strategy, the carousel has to take the shortest route among the different routes that change direction at most once, after retrieving m items. The authors showed that the average travel time is shorter for the m-step strategy compared to the nearest item strategy proposed by Litvak and Adan (2001).

When several orders have to be picked, the question of order sequencing was raised. Lee and Kuo (2008) provided an algorithm for optimizing the sequence of items and orders with the objective of minimizing the total picking cost. This cost was assumed to be dependent only on the distance the picker has to travel to take the items from and place the items in bins, and on the distance travelled by the carousel to look for trays.

2.1.3. AVS/RS

A pioneer process for AVS/RS was proposed by Malmborg (2002). In that study, the author developed an analytical model to evaluate vehicle utilization and cycle time. Kuo, Krishnamurthy, and Malmborg (2007) proposed an analytical model to estimate resource utilization and system cost using twelve scenarios, which depend on vehicle and lift travel times and different probability distributions. Kuo, Krishnamurthy, and Malmborg (2008) extended this model by considering an alternative class-based storage assignment. Marchet, Melacini, Perotti, and Tappia (2012) studied system performances in terms of transaction cycle time and waiting time using an open queueing network approach. They used a simulation to validate their approach.

Some works focused on the design of AVS/RS. For instance, Fukunari and Malmborg (2008) extended the work of Malmborg (2002) and simulated 15 cases with different characteristics of the AVS/RS. They showed that the optimal AVS/RS configurations tend to use more and shorter aisles in order to optimize the movement pattern of vehicles, and shorter racks to avoid inefficiencies associated with vertical travel. Ekren and Heragu (2011) varied the number of lifts and vehicles and studied system performances in terms of utilization and cycle times. They used a simulation-based regression analysis in order to identify the factors impacting AVS/RS performances. Marchet, Melacini, Perotti, and Tappia (2013) investigated the cost and performance trade-off, for AVS/RS new design solutions. They considered several design scenarios and simulated costs and performances in terms of cycle time and flow time for warehouse managers.

2.1.4. SBS/RS

SBS/RS are relatively new and operate generally in mini load warehouses (Malmborg, 2002). This type of AVS/RS, closer to the VLM systems, normally uses aisle and tier-captive shuttles for horizontal movement, while vertical movement is achieved by elevator lifting tables. Lerher, Edl, and Rosi (2014) and Lerher (2013) studied energy efficiency and the impact of CO_2 reduction in SBS/RS for several design scenarios, together with a given number of units to load, in the case of the dual command cycle (i.e., the machine

can perform one storage and one retrieval per cycle). Carlo and Vis (2012) focused on the SBS/RS scheduling issue with a two non-passing lifting system. The authors evaluated throughput improvement and total handling times for different strategies. Lerher, Ekren, Sari, and Rosi (2015) and Lerher (2016) proposed analytical travel time models for SBS/RS. In these papers, the mean travel times in the case of the single command cycle (i.e., the machine can perform one storage or one retrieval per cycle) or the dual command cycle are provided, in addition to the machine characteristics (such as number of aisles, storage length, and number of tiers). Lerher et al. (2016) evaluated throughput performance for SBS/RS in the case of both the single command cycle and the dual command cycle. Recently, Lerher, Borovinsek, Ficko, and Palcic (2017) have developed a simulation study to identify the factors that affect the performance of SBS/RS. In this paper, both single command cycle and dual command cycle were investigated. The authors came out with recommendations for designing efficiently SBS/RS such as using long racks and small numbers of tiers. The results showed that SBS/RS with an adequate design are effective in reducing the mean cycle time.

2.2. Batching decision in warehousing systems

Following a review of some interesting papers on AS/RS technologies, including those related to VLM systems, we now focus on the papers dealing with batching decisions in warehouses. We remind the reader that our study investigates the problem of batching decisions for warehouses using VLMs. The literature related to the batching problem in warehouses can be classified into two categories: batching for picker to part systems and batching for part to picker systems.

In the classical picker to part systems, in which pickers have to walk down the aisles to retrieve items, order batching is carried out according to a specified layout and a given routing strategy (De Koster et al., 2007). In this case, completion time includes setup time, travel time, search time for identification of order lines, and finally the time required for collecting the items (Tompkins et al., 2010). Many authors (e.g., Bartholdi and Hackman, 2014; Tompkins et al., 2010) argued that travel time is the principal source of waste and is evaluated at about 50% of total completion time. Thus, minimizing travel time was the objective of many papers dealing with picker to part systems, such as Gademann and Van de Velde (2005), Henn, Koch, Doerner, Strauss, and Wäscher (2010), and Bozer and Kile (2008). Other objectives have also been considered in the literature such as minimizing tardiness of orders (Azadnia, Taheri, Ghadimi, Mat Saman, & Wong, 2013; Henn, 2015) or minimizing the penalty cost for tardiness (Bukchin, Khmelnitsky, & Yakuel, 2012). A comprehensive review of picker to part systems was given by De Koster et al. (2007). Further, Henn, Koch, and Wäscher (2012) provided a review of the order batching problem in picker to part systems and described the existing solution approaches.

To our knowledge, in the case of part to picker systems (mostly AS/RS), order batching was considered only with the "person-on-board" AS/RS (Hwang & Lee, 1988). This specific AS/RS uses a picker staying on a rail, cart, or vehicle, which transports him to the storage location of items that have to be collected. Order batching in this case is therefore similar to the case of picker to part systems, since the objective is mostly to minimize picker travel time (Elsayed & Unal, 1989).

We finally conclude that most papers on VLM systems focus on the evaluation and optimization of system performance for a given batch of orders. Our research is in the upstream stage, as we focus on how to group orders in order to improve system performance in terms of completion time. Although the batching decision is very relevant in automated warehouses (as we explained earlier), papers that investigate the batching problem for AS/RS, in general, and for VLM systems, in particular, are missing in the literature. Moreover, on the practical side, we have observed that most companies operating VLM systems use basic tools for batching, such as "first come first served," which are generally very far from optimality. This study tries to close the gap in both theory and practice.

3. Order batching with one VLM

In this section we consider the case of one VLM with one picker. Our objective is to create batches in order to minimize total completion time, which is the time taken by the picker to collect all orders. According to our industrial partners, minimizing total completion time is one of the main objectives of companies using VLMs. In the rest of this section, we will proceed as follows: First, we present the notation and our assumptions. Second, we study and evaluate the completion time for a given set of orders. Third, we develop our optimization model for order batching with one VLM. Based on real data collected from two companies, we finally test our model and compare its performance with the actual batching methods used by these companies.

3.1. Notation and assumptions

We consider the following notation:

- N: total number of customers' orders.
- $\mathbb{N} = \{1, \ldots, N\}$:set of indices for customers' orders.
- S: batch size (i.e., the maximum number of orders to put in a batch). In practice, batch size is generally constant and depends on the number of bins in the picking area.
- B: number of batches to be created. Given the total number of considered orders and the number of orders per batch (both are given input parameters in practice), we have $\lceil B = \frac{N}{S} \rceil$ where $\lceil . \rceil$ is the ceiling function.
- $\mathcal{B} = \{1, \ldots, B\}$: set of indices for all batches.
- K: total number of trays in the VLM.
- $\mathcal{K} = \{1, \ldots, K\}$: set of indices for trays in the VLM.
- K^b: number of trays that have to be visited for batch b (in order to collect the items associated with batch b).
- $Kb = \{1, \ldots, K^b\}$: set of indices for trays that have to be visited for batch b.
- W: picker's total waiting time (sum of all times when the picker is available but the tray from which he has to collect items is not).
- P: total picking time. This corresponds to the time during which the picker is collecting items from the trays (excluding waiting time).
- C: total completion time for all batches, $C = W + P$.
- W^b: picker's waiting time associated with batch b.
- P^b: picking time associated with batch b (excluding waiting time).
- C^b: completion time associated with batch b, $C^b = W^b + P^b$.
- W^b_l: picker's waiting time in front of the *l-th* tray that has to be visited for batch b.
- P^b_l: time required to pick items from the *l-th* tray associated with batch b (excluding waiting time).
- C^b_l: completion time for the *l-th* tray associated with batch b, $C^b_l = W^b_l + P^b_l$.
- L^b_l: number of order lines that have to be collected on the *l-th* tray visited for batch b.
- p: picking time of one order line from a given tray when this tray is available. We remind the reader that an order line corresponds to a given quantity of a given product associated with a given customer order. We assume that p is the same for any order line (namely, p does not depend on the order line to be

collected). In practice, p may slightly vary as a function of the number of items (quantity) in the order line and the type of operations to be performed after collecting items (e.g., labeling or scanning items before putting them into the bins). These factors are intrinsic to each warehouse and are difficult to estimate in practice. Furthermore, according to KLS Logistic Systems, a French editor of warehouse management systems, the variation in the value of p between the different order lines in a given warehouse is not very significant for most of their customers operating VLMs, which motivated our assumption of a constant value for p.

It is also important to highlight that the picking time from a given tray is equal to the sum of picking times of all order lines that have to be collected from this tray. Picking time per tray is, of course, variable but this variability mostly depends on the number of order lines to be collected and not on the interaction between the picking operations of these lines. In our models, we take this variation into account as we study the case where the picking time per tray is equal to the picking time of one order line multiplied by the number of lines that have to be collected.

- r: time required by the VLM to store the current tray and to extract the next one. We assume the value of r to be the same for all trays (i.e., it does not depend on the tray in question). This is a realistic assumption since the vertical moving speed of the VLM is high (up to 2 meters per second according to Kardex Remstar, a VLM manufacturer), whereas the horizontal speed is much lower. This implies that the time required to release a tray consists essentially of the time for the horizontal movement, thus justifying our assumption of a constant r. We also verified this assumption by observing real operating conditions in some warehouses using VLMs. We discovered that tray release time does not significantly vary from one tray to another.

To simplify the presentation of the problem, we first address the situation where the picking time is considered negligible compared to the time required by the VLM to store a tray and extract the next one. We then turn to a more complex and realistic situation where the picking time from the *l-th* tray visited depends on the number of lines that have to be collected from this tray, namely $P^b_l = L^b_l * p$, $\forall l \in \mathcal{B}$ and $\forall\ l \in Kb$.

We also assume throughout the paper that the process for collecting a given batch starts by waiting for the first tray. Finally, we recall that total completion time C is given by Eq. (1).

$$C = W + P = \sum_{b=1}^{B} W^b + P^b = \sum_{b=1}^{B} \sum_{l \in \mathcal{K}b} \left(W^b_l + P^b_l\right) \tag{1}$$

As we explained earlier in this paper, the completion time is the sum of the waiting time (when the picker waits for the trays in front of the machine) and picking time (when the picker effectively collects items from the trays).

Based on the definitions above, we have the following relations:

$$C = \sum_{b=1}^{B} C^b \tag{2}$$

$$W = \sum_{b=1}^{B} W^b = \sum_{b=1}^{B} \sum_{l \in \mathcal{K}b} W^b_l \tag{3}$$

$$P = \sum_{b=1}^{B} P^b = \sum_{b=1}^{B} \sum_{l \in \mathcal{K}b} P^b_l \tag{4}$$

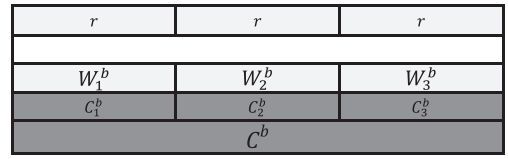

Fig. 3. Illustration of completion time with one VLM when per-tray picking time is negligible.

3.2. Evaluation of completion time with one VLM

Using the above notation and assumptions, the purpose of this section is to evaluate the completion time C of B batches in the case of one VLM.

Proposition 1. *In the case of one VLM, whatever the assumption for the per-tray picking time (either negligible or proportional to the number of order lines), minimizing the total completion time is equivalent to minimizing the number of trays visited for the different batches.*

In the case of negligible per-tray picking time, the total completion time is given by: $C = r * \sum_{b=1}^{B} K^b$.

In the case of per-tray picking time proportional to the number of lines, the total completion time is given by: $C = r * \sum_{b=1}^{B} K^b + p * \sum_{b=1}^{B} \sum_{l \in \mathcal{K}b} L^b_l$.

To make this easier to understand, we first provide an example of each situation. The proof of Proposition 1 is presented in appendix A. In the examples below, we illustrate the calculation of the completion time for only one batch.

Example 1. Completion time with one VLM when the per-tray picking time is negligible

In this case, the picking time from each tray is negligible compared to the other lead times, implying that the picking time is not considered in the calculation of the completion time. We consider the below example with a batch b that requires visiting 3 trays (1, 2 and 3) on the VLM. In Fig. 3, we show how the completion time of batch b is calculated.

The picker begins by waiting in front of the VLM until the first tray is available (tray 1), which takes r units of time. When this first tray is available, the picker collects the items but the picking time in this case is negligible. When he finishes, the first tray is stored and the next one is extracted, which also takes r units of time. Meanwhile, the picker has to wait. When the second tray (in this case, tray 2) becomes available, the picker collects the items in negligible time. The second tray is then stored and the third one is extracted, which adds r more units of waiting time. Finally, the picker collects the items from the third tray, thus completing the collecting process for batch b. Hence, we have $C^b = W^b = \sum_{l \in \mathcal{K}b} W^b_l = 3 * r$.

Example 2. Completion time with one VLM when the per-tray picking time is proportional to the number of lines

We now consider a more realistic case where the picking time for a given tray is proportional to the number of lines (i.e., number of products) to be collected from this tray. We consider a batch b that has to visit 3 trays on the VLM where $L^b_1 = 2$, $L^b_2 = 1$, and $L^b{}_3 = 3$. We remind the reader that L^b_l is the number of order lines that must be collected on the *lth* tray visited for batch b. In Fig. 4, we illustrate how the completion time is obtained.

The picker begins by waiting for the first tray (which requires r units of time), before collecting the two lines from this first tray (which requires $P^b_1 = 2 * p$ units of time). The first tray is then stored and the next one is extracted. Meanwhile, the picker has to wait for r units of time. When the second tray becomes available, the picker collects one line (which requires $P^b_2 = p$ units of

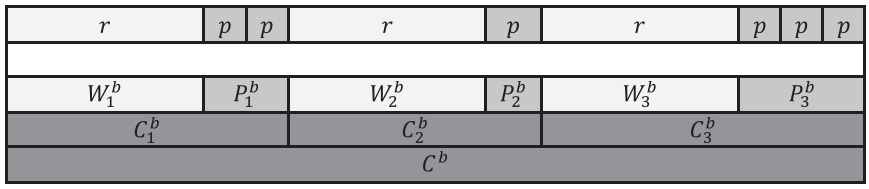

Fig. 4. Illustration of completion time with one VLM when per-tray picking time is proportional to the number of lines.

time). He then waits for the third tray (which requires r units of time) and collects 3 order lines from this tray when it becomes available (which requires $P_3^b = 3 * p$ units of time). This completes the collecting process for batch b. Hence, the completion time for this batch is: $C^b = W^b + P^b = \sum_{l \in \mathcal{K}b}(W_l^b + P_l^b) = r + 2 * p + r + p + r + 3 * p$.

3.3. Order batching with one VLM: optimization model and experiments

In this section, we develop the optimization model for the batching strategy with one VLM. Then, we present and discuss the results of the experiments, which serve to validate the model and to evaluate its performance.

3.3.1. Optimization model

The main decision of our model is how to build the different batches from the set of customers' orders. The decision variables are as follows:

- $x_{nb} = 1$ if order n is included in batch b; 0 otherwise.
- $T_{kb} = 1$ if tray k is visited for batch b; 0 otherwise.

We have demonstrated that minimizing the total completion time is equivalent to minimizing the total number of trays visited (for all batches), given by $\sum_{b=1}^{B} K^b$. In this case, it is important to note that $K^b = \sum_{k=1}^{K} T_{kb}$. Consequently, the objective function of the batching model with one VLM is as follows:

$$Minimize \sum_{b=1}^{B}\sum_{k=1}^{K} T_{kb} \tag{5}$$

The constraints of the model are presented hereafter.

$$\sum_{n=1}^{N} x_{nb} \leq S, \ \forall\ b \in \mathcal{B} \tag{6}$$

$$\sum_{b=1}^{B} x_{nb} = 1, \ \forall\ n \in \mathbb{N} \tag{7}$$

$$T_{kb} = \sum_{n=1}^{N} x_{nb} * U_{kn}, \ \forall\ b \in \mathcal{B}, \ \forall\ k \in \mathcal{K} \tag{8}$$

$$x_{nb} \in \{0; 1\}, \ \forall\ n \in \mathbb{N}, \ \forall\ b \in \mathcal{B}$$

$$T_{kb} \in \{0; 1\}, \ \forall\ k \in \mathcal{K}, \ \forall\ b \in \mathcal{B}$$

Constraint (6) imposes the restriction that the maximum number of orders per batch is equal to S. Constraint (7) ensures that each customer's order is included in one and only one batch. In constraint (8), we use the input parameter U_{kn}, which is equal to 1 if order n contains items that are stored in tray k, and 0 otherwise. Constraint (8) serves to compute the variable T_{kb}. Indeed, a batch b visits tray k if at least one order in this batch requires visiting tray k.

3.3.2. Experiments with real data

We conducted numerical experiments, using the commercial optimization software Cplex to evaluate the gain obtained by using our model in practice. Thanks to our collaboration with KLS Logistic Systems, a French editor of warehouse management systems, we were able to use real data from two companies operating in two different sectors: a hospital in France and a manufacturing company based in Switzerland. The French hospital warehouse provides medical products, such as medicine, to fulfil the hospitals' needs. The Swiss company sells precision measuring instruments, such as gauges, calibration equipment, calipers, and micrometers.

The data, extracted from the databases of these companies, correspond to customers' orders received by the companies over several days. The main difference between the data from the two companies is the average number of orders per day and the average number of order lines per order. We observed that there were fewer orders per day and fewer order lines per order for the French hospital. In both companies, the actual batching strategy is that a batch includes the orders according to their preparation due date until the batch is full (S is reached) and then the process is continued by filling the next batch, until all orders have been allocated to batches. The order management rules of these companies are in line with the context of our models, in which we assume that batches are created from a set of orders that are known in advance. In the companies considered, orders that must be shipped in the afternoon are collected in the morning, and the morning orders are collected in the afternoon of the previous day. Moreover, the picker's work consists of several activities throughout the day: picking items from the VLM, loading goods into the trucks in the shipment zone, and placing items in stock. It is a fact that minimizing completion time means allocating more time to other activities. This is another argument that shows the interest in reducing completion time of order picking in the two companies.

The comparison between our model and the actual batching method in the companies considered will be based on two criteria: the total number of trays visited and the total completion time. Here we ask the reader to note that although minimizing the number of trays visited has been demonstrated to be equivalent to minimizing the completion time (in terms of a batching strategy), the performance of our model compared to that of the companies' actual strategies is not necessarily the same for both criteria. We have been informed by KLS Logistic Systems and several warehouse managers that the most important criterion is the completion time. Nevertheless, the number of trays visited is also interesting because it impacts the operating lifetime of VLMs. Indeed, the greater the number of trays visited, the shorter the effective lifetime of the VLM. This comparison was made using the following formula:

$$\text{Gain} = \frac{\text{Nb. trays visited in the existing approach} - \text{Nb. trays visited in our approach}}{\text{Nb. trays visited in the existing approach}}\% \tag{9}$$

In our comparative study, we assume without loss of generality that the time required to store a tray and to extract the next one equals $r = 1$ unit of time. We let CT1 denote the

Table 1
Our model's performance in terms of number of trays visited for the French hospital (one VLM).

Instance	Nb. orders	Nb. order lines	Nb. trays visited Existing approach	Nb. trays visited Our approach	Gain (%)	Computation time (seconds)
Day 1	19	53	40	29	28	1.52
Day 2	20	44	44	33	25	1.63
Day 3	20	54	36	27	25	1.57
Day 4	20	57	47	34	28	1.02
Day 5	35	79	66	43	35	3.52
Day 6	43	81	66	38	42	7.31
Day 7	63	130	121	58	52	109.03

Table 2
Our model's performance in terms of completion time for the French hospital (one VLM).

Instance	CT1 Existing approach	CT1 Cplex	Savings CT1 (%)	CT2 Existing approach	CT2 Cplex	Savings CT2 (%)	CT3 Existing approach	CT3 Cplex	Savings CT3 (%)
Day 1	40	29	28	93	82	12	50.6	39.6	22
Day 2	44	33	25	98	87	11	54.8	43.8	20
Day 3	36	27	25	80	71	11	44.8	35.8	20
Day 4	47	34	28	104	91	13	58.4	45.4	22
Day 5	66	43	35	145	122	16	81.8	58.8	28
Day 6	66	38	42	147	119	19	82.2	54.2	34
Day 7	121	58	52	251	188	25	147.0	84.0	43

completion time for the scenario in which picking time is negligible. For the second scenario, where per-tray picking time is assumed to be proportional to the number of lines, we study two configurations, whose completion times are denoted by CT2 and CT3, respectively. For CT2, we consider an extreme situation where the picking time per order line $p = 1$ (meaning that $p = r$). For CT3, we consider a realistic situation where $p = 0.2$. Indeed, according to KLS Logistic Systems, the real picking time per order line varies between 10% and 30% of the time required to change a tray. This motivated our choice of $p = 0.2$. The comparison of the CTi's ($i = 1,2,3$) was made using the following formula:

$$\text{Savings CTi} = \frac{\text{CTi in the existing approach} - \text{CTi in our approach}}{\text{CTi in the existing approach}}\% \quad (10)$$

In all these cases, we evaluated the gain obtained by using our model instead of the batching approach currently used by the two companies (described above). Our model was solved to optimality using Cplex on an Intel Core i3 dual 2.27 GigaHertz with 3 GigaBytes Ram, running in a 32 bit Professional Windows 7 environment.

3.3.2.1. Case 1: French hospital. Table 1 summarizes the results of the comparison between the actual batching method used by the French hospital and our model, in terms of the number of trays visited. According to the hospital's practice, S was set to 6 orders per batch. In Table 1, seven different instances are considered. Each instance corresponds to a given day and represents the orders received that day. Table 1 also gives the number of orders for each instance and the total number of order lines. We then provide the total number of trays visited, obtained using the company's method (as described above), the total number of trays visited predicted by our model, and the gain resulting from using our model. Finally, we give the computation time required by Cplex to obtain the optimal solution for our model.

Table 2 provides the results of the comparison between the actual batching method used by the French hospital and our model in terms of the completion time (we remind the reader that CT1, CT2, and CT3 correspond, respectively, to negligible p, $p = 1$, and $p = 0.2$). A graphical representation of the results of Table 2 is provided in Fig. 5.

3.3.2.2. Case 2: Swiss company. We now provide the results of the experiments in the case of the Swiss company, specialized in manufacturing precision measuring instruments. Similar to the case of the French hospital, we considered seven instances, each corresponding to orders received on a given day. According to the order picking strategy used by the Swiss company, S in this case is equal to 8.

Table 3 summarizes the results of the comparison between the actual batching method used by the Swiss company and our model in terms of the number of trays visited. We stopped Cplex after one hour if no optimal solution was found. Instances where an optimal solution was not obtained after one hour are marked by "*" in Table 3. For these instances, we took the best feasible solution obtained after one hour.

Table 4 provides the results of the comparison between the actual batching method used by the Swiss company and our model in terms of completion time. The results of Table 4 are illustrated in Fig. 6.

3.3.2.3. General comments on the results. The results of experiments obtained with real data from the two companies show that our batching model leads to significant savings in terms of completion time and number of trays visited compared to the batching strategy currently used by these companies.

In fact, as can be observed in Tables 1 and 3 (for the French hospital and Swiss company), the gain provided by our model in terms of trays visited ranges from 25% to 52% and from 24% to 33%, respectively. Tables 2 and 4 summarize the comparison results according to total completion time, which is the main criterion, for the French hospital and the Swiss company, respectively. For the French hospital, in the CT1 setting, our model leads to savings ranging from 25% to 52%, with average savings of 33%. For CT2, we observe savings ranging from 11% to 25%, with an average of 15%. For the most realistic setting, CT3, we observe that the gain is between 20% and 43%, with an average of 27%. For the Swiss company, the savings in the CT1 setting range from 24% to 33%, with an average of 27%. Regarding CT2, savings range from 9% to 15%, with an average of 11%. Savings for the most realistic case, CT3, are between 18% and 27%, with an average of 22%.

We observe that an increase in unit picking time p leads to a reduction in the gain obtained by using our model in terms of completion time. This is due to the picking time ($p * \sum_{b=1}^{B} \sum_{l \in \mathcal{K}b} L^b_l$), which although included in completion time, is constant, and thus cannot be minimized.

In the case of the French hospital, the solution time with Cplex did not exceed one minute for most instances: only the instance with 63 orders required a relatively long solution time of 109

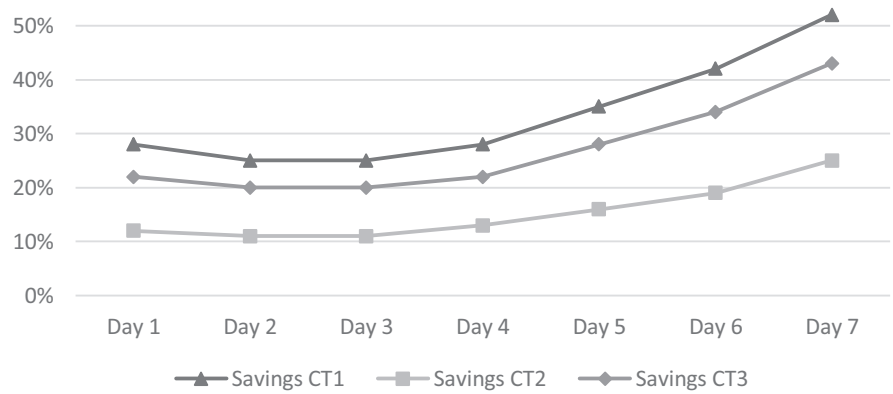

Fig. 5. Completion time savings for the French hospital (one VLM).

Table 3
Our model's performance in terms of number of trays visited for the Swiss company (one VLM).

Instance	Nb. orders	Nb. order lines	Nb. trays visited Existing approach	Nb. trays visited Our approach	Gain (%)	Computation time (seconds)
Day 1	31	269	170	127	25	3.40
Day 2	50	224	169	125	26	78
Day 3	60	307	217	157	28	1190
Day 4	81	352	275	209	24	*
Day 5	103	329	248	166	33	*
Day 6	116	490	327	250	24	*
Day 7	145	364	281	187	33	*

Table 4
Our model's performance in terms of completion time for the Swiss company (one VLM).

Instance	CT1 Existing approach	CT1 Cplex	Savings CT1 (%)	CT 2 Existing approach	CT2 Cplex	Savings CT2 (%)	CT 3 Existing approach	CT3 Cplex	Savings CT3 (%)
Day 1	170	127	25	439	396	10	223.8	180.8	19
Day 2	169	125	26	393	349	11	213.8	169.8	21
Day 3	217	157	28	524	464	11	278.4	218.4	22
Day 4	275	209	24	627	563	10	345.4	281.4	19
Day 5	248	166	33	577	495	14	313.8	231.8	26
Day 6	327	250	24	817	740	9	425.0	348.0	18
Day 7	281	187	33	645	551	15	353.8	259.8	27

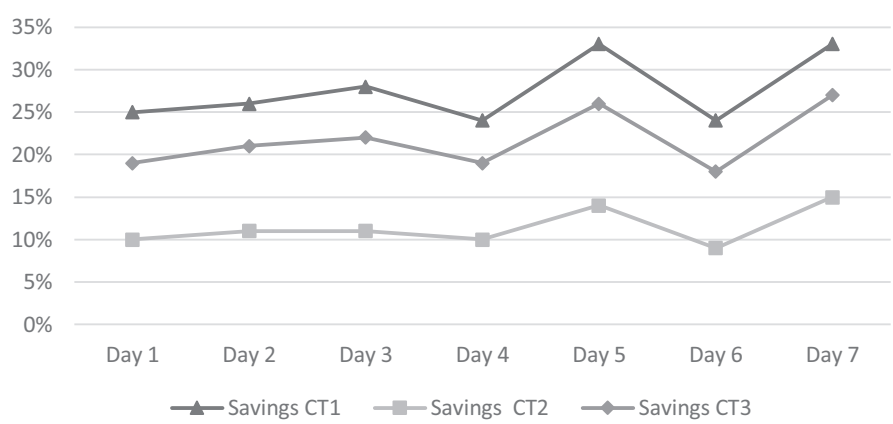

Fig. 6. Completion time savings for the Swiss company (one VLM).

seconds. However, for the Swiss company, we were not able to obtain optimal solutions for all instances due to the large number of orders and order lines. A metaheuristic approach will be proposed in Section 5 for an effective solution in such instances.

3.3.2.4. Effect of the number of orders. To further analyze our results, in this section we study the effect of the number of orders on the performance of our approach (Table 5). For each case (French hospital or Swiss company), we compile two sets. The first set groups the days where the number of orders is less than the median, whereas the second set includes the days with a number of orders greater than the median. When the number of cases is odd, we proceed as follows: if the average number of orders per day is greater than the median value, then the case with the median number of orders is placed in the first set; otherwise, it is placed in the second set. For the French hospital, the median value is 20 and the average number of orders per day is 31. Thus, the first set groups the days where the number of orders is less than

Table 5
Effect of the number of orders on the model's performance (One VLM).

Company	Cluster	Average savings in terms of number of trays visited (%)	Average savings in terms of completion time for the "real" case (CT3) (%)
French hospital	Set 1 (less than or equal to 20 orders per day)	26	21
	Set 2 (more than 20 orders per day)	43	35
Swiss company	Set 1 (less than or equal to 81 orders per day)	26	20
	Set 2 (more than 81 orders per day)	30	24

Table 6
Effect of the number of order lines per order on the model's performance (One VLM).

Company	Cluster	Average savings in terms of number of trays visited (%)	Average savings in terms of completion time for the "real" case (CT3) (%)
French hospital	Set 1 (days with an average number of lines per order less than or equal to 2.26)	39	31
	Set 2 (days with an average number of lines greater than 2.26 lines per order)	27	21
Swiss company	Set 1 (days with an average number of lines per order less than or equal to 4.34)	28	22
	Set 2 (days with an average number of lines greater than 4.34 lines per order)	26	20

or equal to 20 (Days 1–4), whereas the second set includes the days with a number of orders greater than 20 (Days 5–7). For the Swiss company, the median value is 81 and the average number of orders per day is 83. Thus, the first set corresponds to the days where the number of orders is less than or equal to 81 (Days 1–4), and the remaining orders are included in the second set (Days 5–7). For each set, we report in the following table the savings obtained with regard to the number of trays visited and completion time.

The results show that the greater the number of orders considered, the greater the performance of our model. Indeed, a large number of orders increases the probability of having compatible orders (which have many trays in common) that can be placed together in the same batch. This increases the importance of using an optimized batching strategy as proposed in this paper.

3.3.2.5. Effect of the average number of lines per order. In this section, we focus on the number of lines per order. For each company, we create two sets: the days where the average number of lines per order is less than the median value are included in the first set, whereas the second set groups the days where the average number of lines per order is greater than the median. If the number of cases is odd, we use the average value of the average number of lines as a splitting criterion and apply the same splitting rule as described above. Thus, the cases in the first set correspond to the days with an average number of lines per order less than or equal to 2.26 and 4.34 for the French hospital and the Swiss company, respectively. The results are shown in Table 6.

We observe that an increase in the number of lines per order leads to a deterioration in the performance of our model. This result is not intuitive, as we expected that—similarly to the previous case—a larger number of order lines would offer more combination possibilities and consequently a better batching strategy. We analyzed the data considered and deduced that when there are fewer lines per order, the probability of having compatible orders increases (i.e., it becomes more likely to find orders that require visiting the same trays).

Thus, both the number of orders and the number of lines per order have a significant impact on the performance of our batching model. In terms of batching performance, the perfect case would be to have a large number of orders with one line per order (called mono-line orders in practice). In this case, the optimal batching decision would be to group orders that require visiting the same tray.

4. Order batching with several VLMs

We now consider the case of a pod with one picker and several VLMs. The main difference between the case of multiple VLMs and the previous case of one VLM is the ability to work under masked time when there are several VLMs. This allows the picker to collect items from a given VLM while another VLM is extracting a tray, thus resulting in less waiting time for the picker.

Our objective remains the same: developing an optimization model that can be used in practice to minimize completion time. We start by evaluating completion time in the context of several VLMs. We then provide the optimization model and finally present the experiments and numerical results.

4.1. Notation and assumptions

With some adjustments (described below), the notation presented in the previous section continues to be valid in the case of several VLMs.

- V: total number of VLMs.
- $\vartheta = \{1, \ldots, V\}$: set of indices for all VLMs.
- K_v: total number of trays in VLM v .
- $\mathcal{K}_v = \{1, \ldots, K_v\}$: set of indices for trays of VLM v.
- K^b_v: number of trays that have to be visited on VLM v for batch b.
- $\mathcal{K}b_v = \{1, \ldots, K^b_v\}$: set of indices for trays that have to be visited on VLM v for batch b.
- W^b_{lv}: waiting time of the picker in front of the *l-th* tray that has to be visited by VLM v for batch b. This represents the length of time during which the picker waits in front of VLM v before collecting items from the *lth* tray for batch b. Clearly, $W^b_l = \sum_{v=1}^{V} W^b_{lv}$.
- P^b_{lv}: time required to collect items from the *lth* tray of VLM v for batch b. Clearly, $P^b_l = \sum_{v=1}^{V} P^b_{lv}$.

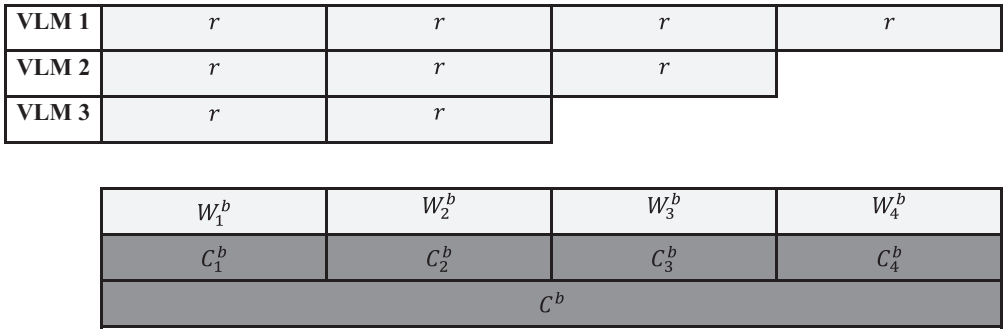

Fig. 7. Illustration of completion time with multiple VLMs when per-tray picking time is negligible.

- L_{lv}^b: number of order lines that have to be collected on the *lth* tray of VLM v for batch b.

In what follows, we state the main assumptions. We first restrict the sequence adopted by the picker to visit the different VLMs: For each batch b, the picker begins with the VLM with the largest number of trays to visit for this batch. The second VLM visited by the picker is the one with the second largest number of trays to visit for batch b, and so on, up to the last VLM (with the smallest number of trays to visit for batch b). After the first round, the picker starts another round with the same sequence of VLMs, and so on, until all items associated with batch b are collected. Moreover, we assume that the time required by the picker to walk from one VLM to another does not need to be considered, which is a realistic assumption given the small size of pods per picker in most real-world situations.

Without loss of generality, for each batch b, we rename the different VLMs such that VLM 1 denotes the VLM with the largest number of trays to visit for batch b, VLM 2 is that with the second largest number of trays to visit for batch b, and so on. According to this notation, we have: $K^b{}_1 \geq K^b{}_2 \geq \ldots \geq K^b{}_V \ \forall\ b \in \mathcal{B}$. Note that the first VLM for batch b is not necessarily the first VLM for another batch b'.

Finally, we remind the reader that we investigate two different cases: (1) per-tray picking time is considered negligible compared to the time required by the VLM to store a tray and extract the next one, and (2) picking time on the *l-th* visited tray of VLM v depends on the number of lines that have to be collected from this tray: $P_{lv}^b = L_{lv}^b * p \ \forall\ v \in \vartheta,\ \forall\ b \in \mathcal{B}$, and $\forall\ l \in \mathcal{K}b_v$.

4.2. Evaluation of completion time with multiple VLMs

The purpose of this section is to evaluate the total completion time C for B batches in the case of several VLMs for the two situations considered: negligible per-tray picking time and per-tray picking time proportional to the number of lines.

4.2.1. Completion time with multiple VLMs and negligible per-tray picking time

To facilitate understanding, we first provide an illustrative example with one batch and then develop the general result in Proposition 2. We still assume that the picker starts by waiting for the first tray.

Example 3. Multiple VLMs with negligible per-tray picking time

We consider a example with 3 VLMs and a batch b having to visit the following number of trays on each of VLM: $K^b{}_1 = 4$, $K^b{}_2 = 3$, and $K^b{}_3 = 2$. In Fig. 7, we show how the completion time for batch b is calculated. The picker begins the first round by waiting for the first tray of VLM 1 to become available, which requires r units of time. We remind the reader that the different VLMs take the same time to extract a tray, implying that the first trays that have to be visited on VLMs 1, 2, and 3 will be available at the same time. The picker collects items from VLM 1 and then moves to VLMs 2 and 3. However, the picking time is negligible here, implying that the duration of the first round is equal to r.

The picker then begins the second round by waiting for the second tray of VLM 1. After r units of waiting time, he collects the items and moves to the second trays of VLMs 2 and 3, which will be immediately available (no more waiting time is required in front of VLMs 2 and 3). In the third round, the picker waits for the third tray of VLM 1, which requires r more units of waiting time, and then moves to the third tray of VLM 2 (VLM 3 is not visited in the third round). Finally, the picker waits for the last visited tray on VLM 1 (VLMs 2 and 3 are not visited in the fourth round). Consequently, the completion time for this batch is equal to the sum of the waiting times in front of the first VLM, which is equal to $4 * r$ (we remind the reader that the first VLM for a given batch is the one visiting the greatest number of trays).

Proposition 2. *In the case of multiple VLMs with negligible per-tray picking time, the total completion time is* $C = r * \sum_{b=1}^{B} K^b{}_1$. *In this case, minimizing the total completion time is equivalent to minimizing the sum of the number of trays visited by the first VLM of each batch,* i.e., *the VLM visiting the greatest number of trays for the given batch.*

The proof is given in appendix B.

4.2.2. Completion time with multiple VLMs and per-tray picking time proportional to the number of lines

We now consider the most complex case, where we have several VLMs and the picking time for a given tray is proportional to the number of lines that have to be collected from this tray. We provide an illustrative example with one batch and then develop the general result in Proposition 3.

Example 4. Multiple VLMs with per-tray picking time proportional to the number of lines

In this case we consider 3 VLMs and the following characteristics for a given batch b:

$K^b{}_1 = 4,\ L^b{}_{1\ 1} = 1,\ L^b{}_{2\ 1} = 1,\ L^b{}_{3\ 1} = 1,\ L^b{}_{41} = 1.$
$K^b{}_2 = 3,\ \ L^b{}_{1\ 2} = 2,\ L^b{}_{2\ 2} = 1,\ L^b{}_{3\ 2} = 2.$
$K^b{}_3 = 2,\ \ L^b{}_{1\ 3} = 2,\ \ L^b{}_{2\ 3} = 2.$

In Fig. 8, we show how the completion time for batch b is calculated. The waiting time for the first tray of the first VLM is $W^b{}_{1\ 1} = r$. After this waiting time, the picker collects 1 line from the tray (since $L^b{}_{1\ 1} = 1$), which takes $P^b{}_{1\ 1} = p$ units of time. The picker then moves to the second VLM. The waiting time for the first tray of the second VLM is $W^b{}_{1\ 2} = 0$ since all VLMs extract their first trays at the same time. The picker collects 2 lines from the first tray of the second VLM (since $L^b{}_{1\ 2} = 2$) within $P^b{}_{1\ 2} =$

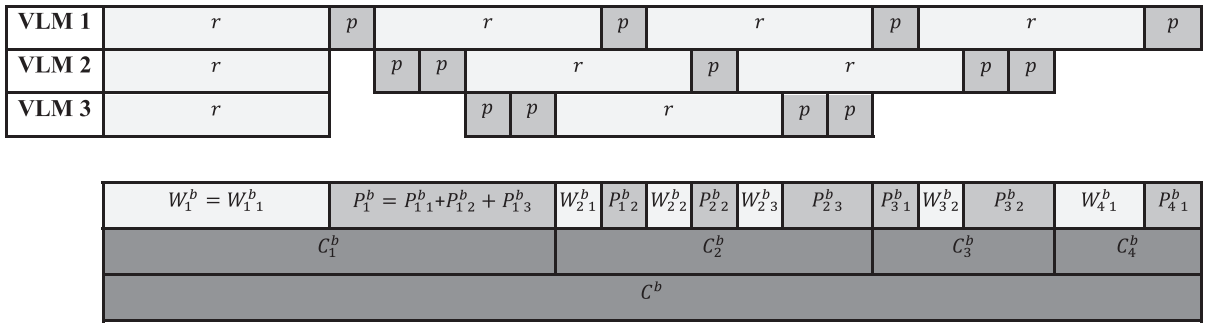

Fig. 8. Illustration of completion time with multiple VLMs when per-tray picking time is proportional to the number of lines.

$2 * p$ units of time, and moves to the third VLM. The waiting time in front of the first tray of VLM 3 is $W^b_{1\,3} = 0$. The picker collects 2 lines (since $L^b_{1\,3} = 2$), which takes $P^b_{1\,3} = 2 * p$ units of time.

After this first round, the picker returns to the first VLM and starts the second round by waiting for the second tray of VLM 1. This waiting time $W^b_{2\,1}$, if any, is the difference between the time required to extract the second tray of VLM 1 (namely, r time units) and the time elapsed between the moment the picker finishes collecting items from the first tray of VLM 1 and the moment he returns to VLM 1 to start the second round (namely, $W^b_{1\,2} + P^b_{1\,2} + W^b_{1\,3} + P^b_{1\,3} = 0 + 2{*}p + 0 + 2{*}p$). If this difference is negative, then no waiting time is required in front of the second tray of VLM 1. Consequently, $W^b_{2\,1} = max(r - W^b_{1\,2} - P^b_{1\,2} - W^b_{1\,3} - P^b_{1\,3}; 0) = max(r - 4 * p; 0)$. After this waiting time, the picker collects 1 item (since $L^b_{2\,1} = 1$), which takes $P^b_{21} = p$ units of time, and moves to VLM 2.

The waiting time $W^b_{2\,2}$ for the second tray of the second VLM, if any, is the difference between the time required to extract the second tray of VLM 2 (namely, r) and the time elapsed between the moment the picker finishes collecting items from the first tray of VLM 2 and the moment he returns to VLM 2 to start the second round (namely, $W^b_{1\,3} + P^b_{1\,3} + W^b_{2\,1} + P^b_{21} = 0 + 2 * p + max(r - 4 * p; 0) + p$). If this difference is negative, then no waiting time is required in front of the second tray of VLM 2. Hence, $W^b_{2\,2} = max(r - W^b_{1\,3} - P^b_{1\,3} - W^b_{2\,1} - P^b_{21}; 0) = max(r - max(r - 4 * p; 0) - 3 * p; 0)$. After this waiting time, the picker collects 1 item (since $L^b_{2\,2}$=1), which takes $P^b_{2\,2} = p$ units of time, and moves to VLM 3.

Similarly, the waiting time in front of the second tray of VLM 3 is $W^b_{2\,3} = max(r - W^b_{2\,1} - P^b_{2\,1} - W^b_{2\,2} - P^b_{2\,2}; 0)$ and the picking time for this tray is $P^b_{2\,3} = 2 * p$ units of time. This finishes the second round and the picker can now start the third round. For the third tray of VLM 1, we have $W^b_{3\,1} = max(r - W^b_{2\,2} - P^b_{2\,2} - W^b_{2\,3} - P^b_{2\,3}; 0)$ and $P^b_{3\,1} = p$ units of time. We then have $W^b_{3\,2} = max(r - W^b_{2\,3} - P^b_{2\,3} - W^b_{3\,1} - P^b_{3\,1}; 0)$ and $P^b_{3\,2} = 2 * p$ units of time for the third tray of VLM 2. We ask the reader to note that no items are required from VLM 3 in the third round (since $K^b_3 = 2$). Therefore, the picker returns directly to VLM 1 and starts the fourth round.

The waiting time in front of the fourth tray of VLM 1 is $W^b_{4\,1} = max(r - W^b_{3\,2} - P^b_{3\,2}; 0)$. We ask the reader to note in this case that the waiting and picking times on VLM 3 in the third round are not included in the calculation of $W^b_{4\,1}$ since VLM 3 is not visited in the third round. Finally, the picker collects items from the fourth tray of VLM 1, which takes $P^b_{4\,1} = p$ units of time. Since no more items are required, the process of collecting items for batch b is completed, and we obtain the completion time shown in Fig. 8.

Proposition 3. *In the case of multiple VLMs with per-tray picking time proportional to the number of lines, the total completion time is given by* $C = \sum_{b=1}^{B} \sum_{l \in \mathcal{K}b_v} \sum_{v=1}^{V} (P^b_{l\,v} + W^b_{l\,v})$, *where*

$$W^b_{lv} = \max\left(r - \left(\sum_{i=1}^{v-1} \left(L^b_{l\,i} * p + W^b_{l\,i} \right) + \sum_{i=v+1}^{V} \left(L^b_{l-1\,i} * p + W^b_{l-1\,i} \right) \right); 0 \right) \ \forall 1 < v < V,$$

$$W^b_{l\,1} = \max\left(r - \sum_{i=2}^{V} \left(L^b_{l-1\,i} * p + W^b_{l-1\,i} \right); 0 \right),$$

$$W^b_{lv} = \max\left(r - \sum_{i=1}^{V-1} \left(L^b_{l\,i} * p + W^b_{l\,i} \right); 0 \right), \text{ and } P^b_{lv} = L^b_{lv} * p.$$

The proof is presented in appendix C.

4.3. Order batching with multiple VLMs: optimization model and experiments

We now develop the optimization model for order batching with several VLMs and then evaluate its performance on real data. We remind the reader that our objective is to reduce completion time. The expression of completion time for multiple VLMs, given in Proposition 3, is very complex. We ask the reader to also note that the calculation of completion time for a given batch, presented in Example 4, assumes that the batch characteristics are known in advance, whereas these characteristics must be the result of the batching strategy in the batching model. Thus, integrating the completion time function of Proposition 3 as an objective function for our constrained optimization model leads to a very complex nonlinear model that does not have a closed-form solution. Furthermore, such a model will not be understandable to managers, and thus cannot be used in practice.

Therefore, we decided to use an alternative objective function, which is the sum of the number of trays visited by the first VLM of each batch (we remind the reader that the first VLM of a given batch is the one that requires visiting the largest number of trays). This choice is motivated by three reasons. First, we demonstrated in the case of multiple VLMs with negligible per-tray picking time that minimizing the total completion time is equivalent to minimizing the sum of the number of trays visited by the first VLM of each batch. Thus, it makes sense to start by investigating this same objective function when per-tray picking time is proportional to the number of lines. Second, the expression for completion time in Proposition 3 includes the number of trays visited by the first VLM of each batch. Third, numerical experiments showed a positive correlation between minimizing the sum of the number of

trays visited by the first VLM of each batch and minimizing the completion time given in Proposition 3.

Thus, we will use (*Minimize* $\sum_{b=1}^{B} K^b{}_1$) as an objective function for our batching model for multiple VLMs. We ask the reader to note once again that $K^b{}_1$ is the result of the batching strategy in our model and, consequently, should be expressed as a function of model variables.

We will then compare the performance of our batching strategy to that of the strategy currently used by the French hospital and the Swiss company with regard to two criteria: the considered objective function and the total completion time (calculated in Proposition 3).

4.3.1. Optimization model

The main purpose of our model is to create batches from a given set of orders. Thus, we consider the following decision variables:

- $x_{nb} = 1$ if order n is included in batch b; 0 otherwise.
- $T_{k\nu b} = 1$ if tray k of VLM ν is visited by batch b; 0 otherwise.

Hence, the number of trays visited by the first VLM for a given batch b is $K^b{}_1 = \max_{\nu\in\vartheta}(\sum_{k=1}^{K} T_{k\ \nu\ b})$. Thus, our objective function is:

$$Minimize \ \sum_{b=1}^{B} \max_{\nu\in\vartheta}\left(\sum_{k=1}^{K} T_{k\ \nu\ b}\right) \tag{11}$$

The model constraints are as follows:

$$\sum_{n=1}^{N} x_{n\ b} \le S, \ \forall\ b \in \mathcal{B} \tag{12}$$

$$\sum_{b=1}^{B} x_{n\ b} = 1, \ \forall\ n \in \mathbb{N} \tag{13}$$

$$T_{k\ \nu\ b} = \sum_{n=1}^{N} x_{n\ b} * U_{k\ \nu\ n}, \ \forall\ b \in \mathcal{B}, \ \forall\ \nu \in \vartheta, \forall\ k \in \mathcal{K}_\nu \tag{14}$$

$$x_{n\ b} \in \{0; 1\}, \forall\ n \in \mathbb{N}, \ \forall\ b \in \mathcal{B}$$

$$T_{k\ \nu\ b} \in \{0; 1\}, \ \forall\ k \in \mathcal{K}, \ \forall\ \nu \in \vartheta, \ \forall\ b \in \mathcal{B}$$

Constraint (12) imposes the restriction of a maximum number of orders per batch. Constraint (13) ensures that each customer's order is included in one and only one batch. Finally, constraint (14) serves to compute variable $T_{k\nu b}$, where the input parameter $U_{k\nu n}$ is equal to 1 if order n contains items that are stored in tray k of VLM ν, and 0 otherwise.

4.3.2. Experiments with real data

We compare the performance of our batching strategy to the strategy currently used by the French hospital and the Swiss company with regard to two criteria: the considered objective function (sum of the number of trays visited by the first VLM of each batch) and the total completion time (calculated in Proposition 3). In both companies, the current batching strategy works by including in a batch the orders according to their preparation due date until the batch is full (i.e., S is reached) and then continuing by filling the next batch until all orders are allocated to batches. We remind the reader that we use real data extracted from the databases of these companies.

As with the case of one VLM, we assume without loss of generality that the time required to store a tray and extract the next one is $r = 1$ unit of time. We also remind the reader that we consider different scenarios for unit picking time. CT1 denotes the completion time for the scenario where the picking time is negligible. For the second scenario where per-tray picking time is proportional to the number of lines, we study the completion times of two configurations: CT2 and CT3. In the case of CT2, we consider an extreme situation where picking time per order line $p = 1$ (i.e., $p = r$). In the case of CT3, we consider a realistic situation where $p = 0.2$ (in practice, p is between 0.1 and 0.3 according to KLS Logistics Systems).

In all these cases, we evaluate the gain obtained by using our model instead of the batching approach currently used by the considered companies (and described above). Our model was solved with Cplex on an Intel Core i3 dual 2.27 GigaHertz with 3 GigaByte Ram, running on the Windows 7 professional 32 bit service pack. We stopped Cplex after one hour if no optimal solution was found. Instances for which an optimal solution was not found after one hour are marked by "*". In this case, we used the best feasible solution obtained.

4.3.2.1. Case 1: French hospital. The pod of the French hospital consists of 2 VLMs with 35 trays each. Therefore, $V = 2$ and $K_1 = K_2 = 35$. According to hospital practices, S was set to 6 orders per batch. Tables 7 and 8 recapitulate the comparison results between the batching method currently used by the French hospital and our model in terms of trays visited and total completion time, respectively. In these tables, 10 different cases were considered, each corresponding to a given day and representing the orders received that day. Fig. 9 illustrates the results of Table 8.

4.3.2.2. Case 2: Swiss company. For the Swiss company, the pod consists of 4 VLMs with 130 trays per VLM. Therefore $V = 4$ and $K_1 = K_2 = K_3 = K_4 = 130$. In this case, S is equal to 8. Tables 9 and 10 show the results of the comparison between our model and the current approach of the firm in terms of the objective function (sum of the number of trays visited by the first VLM of each batch) and total completion time, respectively. In these Tables 11 different cases were considered, each corresponding to a given day and representing the orders received that day. Fig. 10 illustrates the results of Table 10.

4.3.2.3. General comments on the results. Observing the aforementioned tables, we first conclude that our batching model results in considerable savings compared to the batching strategy currently used in the companies considered. Figs. 9 and 10 show that the savings obtained in the more realistic situation, CT3, are significant, thus demonstrating the relevance of our proposed model and its utility in practice. On average, the gains obtained for several VLMs are greater than the gains obtained previously with one VLM. For instance, in a CT3 setting, the average savings of completion time for one VLM are equal to 27% for the French hospital and 22% for the Swiss company. For multiple VLMs, the average savings of completion time are equal to 32% for the French hospital and 24% for the Swiss company. It is a fact that working in masked time is easier with several VLMs, thus emphasizing the advantage of using our batching optimization approach.

Nevertheless, we observe in Table 9 that a significant computation time is required when the number of orders exceeds 60. For those cases, we stopped the solver after one hour, as explained earlier, and took the last feasible solution obtained. We can see that even in these cases, our model outperforms the current batching strategy of the firm with regard to both criteria: trays visited and total completion time. Note that a metaheuristic approach will be developed in Section 5 for an efficient solution in complex cases.

4.3.2.4. Effect of the number of orders. We will now study the impact of the number of orders on the performance of our batching strategy. For each company, we split the considered days of orders

Table 7
Our model's performance with regard to the objective function (sum of the number of trays visited by the first VLM of each batch) for the French hospital (several VLMs).

Instance	Nb. orders	Nb. order lines	Nb. trays visited Existing approach	Nb. trays visited Our approach	Gain (%)	Computation time (seconds)
Day 1	17	32	17	11	35	2,14
Day 2	19	36	19	14	26	2,01
Day 3	36	83	45	30	33	2,09
Day 4	46	65	37	22	41	14
Day 5	47	71	46	29	37	18
Day 6	50	73	48	27	44	21
Day 7	53	119	71	41	42	650
Day 8	60	84	56	31	45	64
Day 9	63	105	69	33	52	1320
Day 10	68	116	82	45	45	*

Table 8
Our model's performance in terms of total completion time for the French hospital (several VLMs).

Instance	CT1 Existing approach	CT1 Cplex	Savings CT1 (%)	CT2 Existing approach	CT2 Cplex	Savings CT2 (%)	CT3 Existing approach	CT3 Cplex	Savings CT3 (%)
Day 1	17	11	35	42	35	17	21,8	15,6	28
Day 2	19	14	26	44	40	9	23,8	19,2	19
Day 3	45	30	33	107	96	10	57,0	42,2	26
Day 4	37	22	41	85	77	9	46,2	32,8	29
Day 5	46	29	37	109	89	18	58,4	41,0	30
Day 6	48	27	44	105	87	17	58,8	38,4	35
Day 7	71	41	42	183	142	22	93,4	61,2	34
Day 8	56	31	45	127	104	18	70,2	45,6	35
Day 9	69	33	52	167	116	31	88,2	48,6	45
Day 10	82	45	45	190	151	21	103,6	66,0	36

Table 9
Our model's performance with regard to the objective function (sum of the number of trays visited by the first VLM of each batch) for the Swiss company (several VLMs).

Instance	Nb. orders	Nb. order lines	Nb. trays visited Existing approach	Nb. trays visited Our approach	Gain (%)	Computation time (seconds)
Day 1	31	269	81	62	23	9
Day 2	45	148	53	39	26	11
Day 3	50	224	76	49	36	30.59
Day 4	60	307	100	73	27	65.61
Day 5	81	352	123	79	36	*
Day 6	103	329	115	67	42	*
Day 7	116	490	143	93	35	*
Day 8	145	364	137	74	46	*
Day 9	160	780	264	186	30	*
Day 10	169	502	187	124	34	*
Day 11	192	1134	204	148	27	*

Table 10
Our model's performance in terms of total completion time for the Swiss company (several VLMs).

Instance	CT1 Existing approach	CT1 Cplex	Savings CT1 (%)	CT2 Existing approach	CT2 Cplex	Savings CT2 (%)	CT3 Existing approach	CT3 Cplex	Savings CT3 (%)
Day 1	81	62	23	293	281	4	104	87	16
Day 2	53	39	26	162	154	5	66.2	52.8	20
Day 3	76	49	35	244	231	5	97	71	27
Day 4	100	73	27	332	315	5	128	101	21
Day 5	123	79	36	388	363	6	154	112	27
Day 6	115	67	42	368	342	7	145	100	31
Day 7	143	93	35	525	506	4	188	143	24
Day 8	137	74	46	424	383	10	176	116	34
Day 9	264	186	30	848	804	5	335	262	22
Day 10	187	124	34	553	524	5	233	173	26
Day 11	204	148	27	685	657	4	264	213	19

into two sets as we did in case of one VLM (based on whether the number of orders per day is greater or less than the median value). The results are shown in the following table. These results are in line with the findings in the case of one VLM. Indeed, an increase in the number of orders leads to an improvement in the performance of our batching model.

4.3.2.5. Effect of the number of order lines per order. Regarding the effect of the number of lines per order, we also create two sets from the days of data of each company, based on whether the number of lines per order on a given day is greater or less than the median value (see the case of one VLM for more details). We give the results in Table 12.

Once again, the results are in line with what we observed in the case of one VLM. Indeed, the fewer the lines per order, the greater

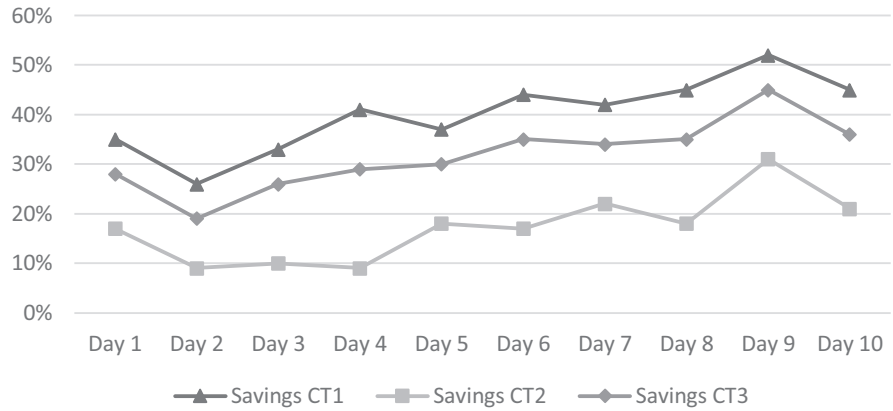

Fig. 9. Completion time savings obtained for the French hospital (several VLMs).

Table 11
Effect of the number of orders on the model's performance (Several VLMs).

Company	Cluster	Average savings in terms of trays visited (%)	Average savings in terms of completion time for the "real" case (CT3) (%)
French hospital	Set 1 (less than or equal to 48.5 orders per day)	34	26
	Set 2 (more than 48.5 orders per day)	46	38
Swiss company	Set 1 (less than or equal to 103 orders per day)	32	24
	Set 2 (more than 103 orders per day)	34	25

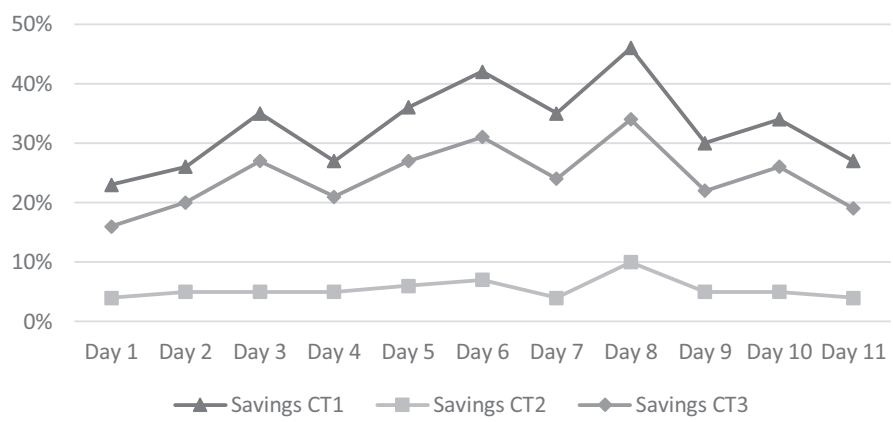

Fig. 10. Completion time savings obtained for the Swiss company (several VLMs).

Table 12
Effect of the number of order lines per order on the model's performance (Several VLMs).

Company	Cluster	Average savings in terms of number of trays visited (%)	Average savings in terms of completion time for the "real" case (CT3) (%)
French hospital	Set 1 (days with 1.69 lines per order or less)	44	35
	Set 2 (days with more than 1.69 lines per order)	36	29
Swiss company	Set 1 (days with 4.34 lines per order or less)	37	29
	Set 2 (days with more than 4.34 lines per order)	27	21

the savings obtained by our batching model (with regard to both criteria: trays visited and completion time). The explanation given in the case of one VLM continues to apply here. Indeed, we observed that when there are fewer lines per order, the probability of having compatible orders increases. In other words, it becomes more likely to find compatible orders (i.e., orders that require visiting the same trays). This gives our model more flexibility to capitalize on masked time.

4.3.2.6. Confirmation of the results with other clusters. We showed that our batching model performs better with an increase in the number of orders or a decrease in the number of lines per order. To demonstrate that this analysis is not specific to the cases con-

Table 13
Comparison results between 4 clusters (Several VLMs).

Days where the number of orders per day is:	Less than or equal to 103		Greater than 103	
Days where the number of lines per order is:	Less than 4.41 (%)	Greater than 4.41 (%)	Less than 4.22 (%)	Greater than to 4.22 (%)
Average savings in terms of trays visited	34.6	29.6	40	30.7
Average savings in terms of completion time for the "real" case (CT3)	26.1	21.3	30	21.7

sidered, we now focus on the case of the Swiss company and consider 4 new clusters as follows.

We first consider 2 groups of days: a first group where the number of orders is less than or equal to the median value (namely, 103) and a second group with a number of orders greater than the median value. Then, based on each group, we compose 2 clusters of days depending on whether the number of lines per order is less or greater than the median value, which is equal to 4.41 for the first group and 4.22 for the second group.

The results, presented in Table 13, confirm our findings. Indeed, on the one hand, we observe that an increase in the number of orders per day leads to a better performance of our batching model, regardless of the average number of lines per order. We observe that, with regard to completion time, savings increase from 26.1% to 30% (for cases with a small number of lines per order) and from 21.3% to 21.7% (for cases with a large number of lines per order). On the other hand, a decrease in the number of lines per order results in an improvement in our batching performance, regardless of the average number of orders per day. With regard to completion time, savings increase from 21.3% to 26.1% (for cases with a small number of orders per day) and from 21.7% to 30% (for cases with a large number of orders per day).

5. A metaheuristic approach for complex cases

We observed in the previous section that significant computation time is required when the number of orders exceeds 60 in the case of multiple VLMs. As our main objective is to develop a batching tool that will be used in practice, we need to guarantee that a good (near optimal) solution can always be obtained within a reasonable amount of time.

For KLS Logistic Systems, the computation time for batching must not be more than five minutes. Therefore, we developed a heuristic approach that could be used by the company for complex cases (when Cplex fails to find the optimal solution quickly). We tested different metaheuristics (Simulated Annealing, Tabu Search, and Genetic Algorithms). The best results were obtained with Simulated Annealing (SA). A detailed description of this metaheuristic is available in the literature dealing with order batching (see e.g., Matusiak, De Koster, Kroon, & Saarinen, 2014). In this section, we first present the SA algorithm developed for our batching optimization model. We then conduct experiments and report the performance of the SA metaheuristic. It is important to note that the metaheuristic approach was integrated in the warehouse management system of KLS Logistic Systems, and is currently used by the company.

5.1. The algorithm

The first step in the implementation of the SA metaheuristic is the calibration of parameters. A detailed explanation of the calibration choice is available in Laajili, Lenoble, and Frein (2017). The solution is encoded as shown in the example of Table 14. In this example, we consider $N = 8$ and $S = 3$ (namely, 8 orders and a maximum of 3 orders per batch). We observe that batch 1 includes orders 3 and 4, batch 2 includes orders 1, 5 and 6, and batch 3 is composed of orders 2, 7 and 8.

Table 14
Solution structure.

Batch number	2	3	1	1	2	2	3	3
Order number	1	2	3	4	5	6	7	8

Regarding the generation of the initial solution, we chose the current batching strategy used by the company (this strategy was explained earlier). The algorithm stops when the termination condition is met. In our case, this condition is the limit on computation time, which was set to one and five minutes alternatively, as required by KLS Logistic Systems. We used the SWAP method to carry out the neighborhood moves. This method consists of randomly choosing two different positions in our solution and interchanging their batch numbers (see Table 15). This method is commonly used in metaheuristics. In our model, the SWAP method always provides feasible solutions as it does not change the number of orders per batch. In the following example, we show how the SWAP method works. Order 3 is moved from batch 1 to batch 3, while order 7 is moved from batch 3 to batch 1. The rest remains the same.

The SA algorithm is presented below.

Start
Step 1:
- Choose the initial solution i_0, implying a fitness value $f(i_0)$ (i.e., the value associated with the objective function).
- Set $i^* = i_0$ and $i = i_0$ (i^* is the best solution and i is the current solution).
Step 2:
- Choose the initial value of the temperature control parameter $T_0 = 0.1$.
- Set the current value of the temperature control parameter T to T_0.
- Choose a final temperature $TF = 0.01$.
- Choose the temperature decrease factor $TDF = 0.9995$.
Step 3:
While {($T > TF$) and (limit computation time not reached)} do {
- Use the SWAP method to find neighbor solutions (neighborhood size is 3 times the number of orders).
- For each neighbor solution i', we calculate $\Delta f = f(i') - f(i)$.
- If (there is at least one neighbor solution such as $\Delta f < 0$),
Then replace the current solution i by the solution with the smallest value of Δf.
Otherwise {
- For each neighbor solution, set a random number $r \in]0, 1[$.
-If ($r < \exp(\frac{-\Delta f}{T})$) Then keep this solution in the pool.
Otherwise remove this solution from the pool.
- Replace the current solution i by the best neighbor solution from the pool.
}
- If ($f(i) < f(i^*)$) Then replace i^* by i. Otherwise, keep the current value of i^*.
- Set the new value of T to $TDF * T$.
}
End.

5.2. Results

Unlike the cases of the French hospital for which we were always able to find an optimal solution with Cplex (as there are relatively few orders), many cases of the Swiss company were difficult to solve, even after one hour of running Cplex. Thus, we used the data of the Swiss company to evaluate the performance of our SA metaheuristic. The comparison criterion is the sum of the number of trays visited by the first VLM of each batch, which was considered in the objective function of the batching model with multi-

Table 15
SWAP method.

Order number	1	2	3	4	5	6	7	8
Initial solution	2	3	1	1	2	2	3	3
			↑				↑	
Solution after SWAP	2	3	3	1	2	2	1	3

Table 16
Metaheuristic's performance after one minute.

Case	*N*	Existing approach	Cplex (one hour)	SA metaheuristic		
				MFV (one min)	IR (%)	ER (%)
Day 1	31	81	62	62	23	100.00
Day 2	45	53	39	39	26	100.00
Day 3	50	76	49	49,3	35	99.39
Day 4	60	100	73	74	26	98.63
Day 5	81	123	79	78.7	36	100.38
Day 6	103	115	67	72.3	37	92.09
Day 7	116	143	93	93	35	100.00
Day 8	145	137	74	74	46	100.00
Day 9	160	264	186	176.7	33	105.00
Day 10	169	187	124	120.3	36	102.98
Day 11	192	204	148	134.1	34	109.39
Average					33.5	100.71

Table 17
Metaheuristic's performance after five minutes.

Case	*N*	Existing approach	Cplex (one hour)	SA metaheuristic		
				MFV (five min)	IR (%)	ER (%)
Day 1	31	81	62	62	23	100.00
Day 2	45	53	39	39	26	100.00
Day 3	50	76	49	49.3	35	99.39
Day 4	60	100	73	73.7	26	99.04
Day 5	81	123	79	77.7	37	101.65
Day 6	103	115	67	69.3	40	96.57
Day 7	116	143	93	91.7	36	101.40
Day 8	145	137	74	72	47	102.70
Day 9	160	264	186	174.3	34	106.29
Day 10	169	187	124	117.4	37	105.32
Day 11	192	204	148	130.1	36	112.09
Average					34.4	102.22

ple VLMs (see Section 4.3). We compare the performance of our metaheuristic (after one minute and five minutes) to the current batching strategy of the company and to the solutions obtained with Cplex after one hour. Regarding the quality of the solution obtained with the SA metaheuristic, we ran the metaheuristic 10 times for each case considered and took the mean result, called the Mean Fitness Value (MFV). We considered two comparison indicators:

- The Improvement Rate (IR), used to compare the performance of the existing approach used by the company to the performance of our metaheuristic. Clearly, the larger the IR, the greater the gain obtained by using the SA metaheuristic. $\text{IR} = \frac{\text{Result with the existing approach} - \text{MFV}}{\text{Result with the existing approach}}\%$.
- The Efficiency Rate (ER), used to compare the performance of Cplex (after one hour) to our metaheuristic approach. Clearly, when ER is greater than 100%, the metaheuristic yields better results than Cplex. $\text{ER} = 1 - \frac{\text{MFV} - \text{Result of Cplex after 1 h}}{\text{Result of Cplex after 1 h}}\%$.

Tables 16 and 17 present the results of the comparison when we ran our SA metaheuristic for one minute and five minutes, respectively.

From these results, we deduce the following:

- On average, our proposed metaheuristic improves the current batching strategy of the company by 33.5% and 34.4% after one minute and five minutes, respectively.
- On average, if we run our metaheuristic for only one minute, we outperform the Cplex solution obtained after one hour with an average efficiency rate ER = 100.71%. If we run the metaheuristic for five minutes (which is the time limit set by the company), we obtain a better performance compared to Cplex, as ER increases to an average of 102.22%.
- As expected, an increase in the number of orders increases the benefit of using our metaheuristic. Indeed, as we can see in both tables, we generally obtain the best values of ER when we consider cases with a large number of orders. In our experiments, the best values of ER were obtained for Day 11 (which has the largest number of orders). Indeed, ER = 109.39% after one minute and 112.09% after five minutes.
- For cases with a relatively small number of orders, Cplex gives the optimal solution. However, even in this case, the use of our metaheuristic is still interesting. This is because the lowest value of ER is 92.09% and 96.57% after one minute and five minutes, respectively, of running the metaheuristic. Thus, we do not lose very much in terms of the quality of the solution but we do gain in terms of the time required to obtain the solution.

- The results obtained with our metaheuristic after five minutes are generally better than the results for one minute. However, the improvement is not very significant (102.22% vs. 100.71% on average). Therefore, from an industrial perspective, it may be interesting to run the metaheuristic only for one minute.

Overall, we can conclude that the proposed metaheuristic can be efficiently used in practice to solve the batching problem with several VLMs. Indeed, on the one hand, it yields near optimal solutions in a very short time when the optimal solution can be obtained with Cplex and, on the other hand, in only one minute, it improves the solution obtained after one hour of running Cplex (when Cplex fails to reach the optimal solution).

6. Conclusion

This study investigated the problem of order batching in warehouses using the automated storage technology of vertical lift modules. We first evaluated the completion time required to collect a set of customers' orders for a given batching strategy in each of two settings: the picking time per tray is negligible or proportional to the number of items to be collected from the tray. In the case of one VLM, we showed that minimizing total completion time is always equivalent to minimizing the total number of trays visited. For multiple VLMs, we demonstrated that in the case of negligible picking time, minimizing total completion time is equivalent to minimizing the sum of the number of trays visited by the first VLM for each batch (the first VLM for a given batch is the one that requires visiting the greatest number of trays). When the per-tray picking time is proportional to the number of items, we obtained a very complex expression of total completion time and found a positive correlation between this completion time and the sum of the number of trays visited by the first VLM of each batch.

We then developed optimization models for order batching in the cases of one and multiple VLMs. We used as an objective function the minimization of the total number of trays visited and the minimization of the sum of the number of trays visited by the first VLM for each batch in the cases of one and multiple VLMs, respectively. Our models were applied and tested in real-world situations for a French hospital and a Swiss company. We used real data extracted from the databases of these companies to evaluate the performance of our models compared to the batching strategy currently used by these companies. We showed that our models outperform the current batching strategy in the different settings considered. For instance, in the most realistic setting, the average savings in completion time in the case of one VLM were equal to 27% for the French hospital and 22% for the Swiss company. In the case of multiple VLMs, the average savings in completion time were equal to 32% for the French hospital and 26% for the Swiss company. We also showed that our batching model performs better with an increase in the number of orders. We observed that when there are many orders, the probability of having compatible orders (i.e., orders that require visiting the same trays) increases, thus making it more important to use an optimization approach for batching. Moreover, we discovered that an increase in the number of order lines per order leads to a deterioration in the performance of our model. This result was not intuitive as we expected that more order lines would offer more combination possibilities and consequently a better batching strategy. However, when there are more order lines per order, the probability of having compatible orders decreases. In other words, it becomes less likely to find orders that require visiting the same trays.

To reduce the computational time required to solve complex cases, we developed and implemented a metaheuristic approach. We tested the performance of our metaheuristic on real data and showed that when Cplex fails to obtain an optimal solution after one hour, our metaheuristic gives a better solution than the last feasible solution of Cplex after only one minute. For simple cases that can be solved with Cplex, the metaheuristic also performs well as it provides a near-optimal solution in a very short timeframe. We showed that the solutions obtained with our metaheuristic after one minute (five minutes) outperform the solutions of Cplex after one hour with an average efficiency rate of 100.71% (102.22%). Moreover, the metaheuristic improved the current batching strategy of the company by 33.5% and 34.4% after one minute and five minutes, respectively.

Our models assumed that the picking time per order line is deterministic. Although many practitioners argue that this can lead to a good approximation of unit picking time and is justified in many situations, it would be interesting to consider a stochastic unit picking time, especially in the case of several VLMs. This is obviously one of the main branches of our work. This research can also be extended to study the batching strategy of other systems such as carousels, which share many similarities with VLMs. However, our models must be adapted, as the time required to release a tray from a carousel is highly correlated with the sequence of retrieving the different items, whereas this time is roughly the same for all trays in a VLM.

Appendix A. Proof of Proposition 1

First, recall that the total completion time is given by $C = \sum_{b=1}^{B} C^b$ and $C^b = W^b + P^b$.

In the case of negligible per-tray picking time, $P_l^b = 0 \ \forall \ l \in K^b$ and, consequently, $P^b = 0$. Therefore, $C^b = W^b = \sum_{l \in K^b} W_l^b = K^b * r$. The total completion time is then given by: $C = \sum_{b=1}^{B} K^b * r$. Recall that r is constant. Consequently, $C = r * \sum_{b=1}^{B} K^b$. Moreover, we deduce that when the per-tray picking time is negligible, minimizing the total completion time C is equivalent to minimizing the total number of trays visited, given by $(\sum_{b=1}^{B} K^b)$.

If the per-tray picking time is proportional to the number of lines, we have $C^b = W^b + P^b = \sum_{l \in \mathcal{K}b}(W_l^b + P_l^b) = K^b * r + \sum_{l \in \mathcal{K}b} L_l^b * p$. Therefore, the completion time for all batches is given by: $C = \sum_{b=1}^{B}(K^b * r + \sum_{l \in \mathcal{K}b} L_l^b * p) = r * \sum_{b=1}^{B} K^b + p * \sum_{b=1}^{B} \sum_{l \in \mathcal{K}b} L_l^b$, since r and p are constant. In this case, it is important to note that $\sum_{b=1}^{B} \sum_{l \in \mathcal{K}b} L_l^b$ is the total number of collected lines, which is a constant input parameter (it does not depend on the batching decision). Hence, when the per-tray picking time is proportional to the number of lines, minimizing the total completion time is also equivalent to minimizing the total number of trays visited, given by $(\sum_{b=1}^{B} K^b)$.

Appendix B. Proof of proposition 2

In case of negligible per-tray picking time (*i.e.*, $P_l^b \cong 0$ and therefore $P^b \cong 0$), we have for a given batch b, $C^b = W^b = \sum_{v \in \vartheta} \sum_{l \in \mathcal{K}b_v} W_{l\,v}^b = \sum_{l \in \mathcal{K}b_1} W_{l\,1}^b$, since waiting time occurs only in front of the first VLM (the one visiting the greatest number of trays). Thus, the total completion time for batch b is $C^b = \sum_{l \in \mathcal{K}b_1} W_{l\,1}^b = \sum_{l \in \mathcal{K}b_1} r = K_1^b * r$. Therefore, $C = \sum_{b=1}^{B} C_b = \sum_{b=1}^{B}(K_1^b * r) = r * \sum_{b=1}^{B} K_1^b$.

Appendix C. Proof of Proposition 3

By definition, we have $C = \sum_{b=1}^{B} \sum_{l \in \mathcal{K}b_v} \sum_{v=1}^{V} (P_{l\,v}^b + W_{l\,v}^b)$. Given that the per-tray picking time is proportional to the number of lines, the picking time for the lth tray of VLM v for batch b is $P_{lv}^b = L_{lv}^b * p$.

The waiting time W_{lv}^b, if any, is the difference between the time required by VLM v to store tray $(l-1)$ and extract tray l, and the time elapsed between the moment the picker finishes collecting

items on tray ($l-1$) of VLM v and the moment he returns to VLM v to start round l. Hence, if there is a waiting time in front of tray l of VLM v for batch b, then W^b_{lv} is given by the formula (1) - (2) and (3), where:

(1) The time required by VLM v to store tray ($l-1$) and extract tray l. This time is constant for all trays and all VLMs, and is equal to r,
(2) The time spent by the picker for collecting items from VLMs $(v+1)$ to V in round $(l-1)$ and from VLMs 1 to $(v-1)$ in round l, which takes $\sum_{i=v+1}^{V} L^b_{l-1\ i} * p + \sum_{i=1}^{v-1} L^b_{l\ i} * p$,
(3) The waiting times in front of the (l-1)th trays of VLMs $(v+1)$ to V and l^{th} trays of VLMs 1 to $(v-1)$. This is given by $\sum_{i=v+1}^{V} W^b_{l-1\ i} + \sum_{i=1}^{v-1} W^b_{l\ i}$.

If the formula (1)-(2) and (3) leads to a negative value, then there is no waiting time in front of tray l of VLM v for batch b. In this case, we have $W^b_{lv} = 0$. Consequently, $W^b_{lv} = max(r - (\sum_{i=1}^{v-1}(L^b_{l\ i} * p + W^b_{l\ i}) + \sum_{i=v+1}^{V}(L^b_{l-1\ i} * p + W^b_{l-1\ i})); 0)$.

Supplementary materials

Supplementary material associated with this article can be found, in the online version, at doi:10.1016/j.ejor.2017.12.037.

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